

Error bounds for estimators of $f'(x)$

Name	Error bound
Forward difference	$\frac{M_2}{2}h$
Backward difference	$\frac{M_2}{2}h$
Central difference	$\frac{M_3}{6}h^2$

Error bounds for estimators of $\int_a^b f(x)dx$

Name	Formula	Error bound
Trapezoidal	$\frac{h}{2}(f_0 + f_1)$	$\frac{M_2}{12}h^3$
Composite trapezoidal		$\frac{(b-a)M_2}{12}h^2$
Simpson's 1/3	$\frac{h}{3}(f_0 + 4f_1 + f_2)$	$\frac{M_4}{90}h^5$
Composite Simpson's 1/3		$\frac{(b-a)M_4}{180}h^4$

Error bounds for estimators of $f''(x)$

Name	Error bound
Central difference	$\frac{M_4}{12}h^2$

Newton-Raphson

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Midpoint

$$\begin{aligned}\mathbf{s}_0 &= f(\hat{\mathbf{y}}[k], t) \\ \mathbf{y}_1 &= \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_0 \\ \hat{\mathbf{s}} &= f(\mathbf{y}_1, t+h/2)\end{aligned}$$

Euler

$$\hat{\mathbf{s}} = f(\hat{\mathbf{y}}[k], t)$$

RK4

$$\begin{aligned}\mathbf{s}_0 &= f(\hat{\mathbf{y}}[k], t) \\ \mathbf{y}_1 &= \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_0 \\ \mathbf{s}_1 &= f(\mathbf{y}_1, t+h/2) \\ \mathbf{y}_2 &= \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_1 \\ \mathbf{s}_2 &= f(\mathbf{y}_2, t+h/2) \\ \mathbf{y}_3 &= \hat{\mathbf{y}}[k] + h\mathbf{s}_2 \\ \mathbf{s}_3 &= f(\mathbf{y}_3, t+h) \\ \hat{\mathbf{s}} &= (1/6)(\mathbf{s}_0 + 2\mathbf{s}_1 + 2\mathbf{s}_2 + \mathbf{s}_3)\end{aligned}$$

Sample estimates

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\sigma}_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_n)^2}$$

2D

$$\begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

3D x-axis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & \sin(\alpha) \\ 0 & -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

3D y-axis

$$\begin{bmatrix} \cos(\alpha) & 0 & -\sin(\alpha) \\ 0 & 1 & 0 \\ \sin(\alpha) & 0 & \cos(\alpha) \end{bmatrix}$$

3D z-axis

$$\begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Heun

$$\begin{aligned}\mathbf{s}_0 &= f(\hat{\mathbf{y}}[k], t) \\ \mathbf{y}_1 &= \hat{\mathbf{y}}[k] + h\mathbf{s}_0 \\ \mathbf{s}_1 &= f(\mathbf{y}_1, t+h) \\ \hat{\mathbf{s}} &= 0.5(\mathbf{s}_0 + \mathbf{s}_1)\end{aligned}$$

Distributions

$$\mu_X = \int_{\Omega} x p(x) dx$$

$$\sigma_X = \sqrt{E[(X - \mu_X)^2]}$$

$$P(e) = \int_e p(x) dx$$